

JEE MAIN REVISION MASTER: P-BLOCK ELEMENTS SYLLABUS REDUCED

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 254 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	P Block Elements — Applied Cases, Reactions & Traps	157 PYQs	Applied Topics
Very High	P Block Elements — Standard Calculations & Problem Archetypes	80 PYQs	Quantitative Analysis
Very High	P Block Elements — Core Principles & Governing Laws	38 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
P Block Elements — Applied Cases, Reactions & Traps	OH^- , $\text{(CH}_3\text{)}_3\text{Si(OH)}_3$, $\text{C}_2\text{H}_5\text{OH} + \text{PCl}_3 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{A}$
P Block Elements — Standard Calculations & Problem Archetypes	$\text{E}^{\text{+}}/\text{E}$, B_2O_3 , N_2O_5 , SO_3 , P_4O_{10} (acidic oxides), RSiCl_3
P Block Elements — Core Principles & Governing Laws	NH_3 , H_2S , O_7 , OF_2

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 P Block Elements — Applied Cases, Reactions & Traps (Very High Yield — 157 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of P Block Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Find the correct statements related to group 15 hydrides. A. Reducing nature increases from NH_3 to BiH_3 B. Tendency to donate lone..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Correct statements from the following are: A. Nitrogen in oxidation states from +1 to +4 disproportionates in acid medium. B. Nitrogen has t..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I: The second ionization enthalpy of B, Al and Ga is in the order of $B > \text{Al} > \text{Ga}$. Statement II: T..." [Ans: N/A]

3.2 P Block Elements — Standard Calculations & Problem Archetypes (Very High Yield — 80 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in P Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Find the correct statements related to group 15 hydrides. A. Reducing nature increases from NH_3 to BiH_3 B. Tendency to donate lone..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Correct statements from the following are: A. Nitrogen in oxidation states from +1 to +4 disproportionates in acid medium. B. Nitrogen has t..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "A 'p'-block element (E) and hydrogen form a binary cation $(\text{EH}_x)^{+}$, while EH_3 on treatment with K_2HgI_4 in alkaline med..." [Ans: N/A]

3.3 P Block Elements — Core Principles & Governing Laws (Very High Yield — 38 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in P Block Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in P Block Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in P Block Elements PYQs.

Representative Source PYQs:

- [2026 Evening Shift] Q#3 (MCQ): "Given below are two statements : Statement I : The increasing order of boiling point of hydrogen halides is $\text{HCl} < \text{HBr} < \text{HI} < \text{HF}$. Statement I..." [Ans: N/A]
- [2026 Morning Shift] Q#4 (MCQ): "Given below are two statements : Statement I : The covalency of oxygen is generally two but it can exceed upto four. The oxidation state of ..." [Ans: N/A]
- [2025 Morning Shift] Q#3 (MCQ): "Given below are two statements: Statement I: Nitrogen forms oxides with +1 to +5 oxidation states due to the formation of $p\pi-p\pi$ bond w..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#2)

Treatment of a gas 'X' with a freshly prepared ferrous sulphate solution gives a compound 'Y' as a brown ring. The compounds X and Y are.

- (A) NO and $[\text{Fe}(\text{NO})]\text{SO}_4$ **[Correct]**
- (B) NO_2 and $[\text{Fe}(\text{NO}_2)]\text{SO}_4$
- (C) N_2O and $[\text{Fe}(\text{N}_2\text{O})]\text{SO}_4$
- (D) N_2O_4 and $[\text{Fe}(\text{N}_2\text{O}_4)]\text{SO}_4$

Correct Answer: A

Solution Analysis: The reaction described is the well-known brown ring test, which is a confirmatory test for nitrate ions (NO_3^-). Let's understand the chemistry behind it. Understanding the Reactants: The test uses a freshly prepared solution of ferrous sulphate (FeSO_4). It needs to be fresh because the ferrous ion (Fe^{2+}) can be easily oxidized to ferric ion (Fe^{3+}) by the oxygen in the air. The question mentions a gas 'X' is treated with this solution. In a standard brown ring test, this gas is produced in situ (within the reaction mixture). Let's see how. Formation of Gas 'X':

Benchmark Problem 2: JEE Main 2024 (Online) 9th April Evening Shift (Q#1)

Number of oxygen atoms present in chemical formula of fuming sulphuric acid is _____.

Correct Answer: 7

Solution Analysis: Fuming sulfuric acid, often known as oleum, can be represented chemically using the formula $\text{H}_2\text{S}_2\text{O}_7$. This compound is formed by adding excess sulfur trioxide (SO_3) to sulfuric acid (H_2SO_4). The chemical reaction can be represented as follows: $\text{H}_2\text{SO}_4 + \text{SO}_3 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. From the formula $\text{H}_2\text{S}_2\text{O}_7$, to determine the number of oxygen atoms, we look directly at the subscript corresponding to oxygen in the chemical formula. In this case, the subscript is 7, which indicates that there are 7 oxygen atoms present in the molecule of fuming sulfuric acid.

Benchmark Problem 3: JEE Main 2024 (Online) 1st February Evening Shift (Q#2)

Given below are two statements : Statement (I) : SiO_2 and GeO_2 are acidic while SnO and PbO are amphoteric in nature. Statement (II) : Allotropic forms of carbon are due to property of catenation and $p\pi-d\pi$ bond formation. In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Statement I is false but Statement II is true
- (B) Both Statement I and Statement II are false
- (C) Both Statement I and Statement II are true
- (D) Statement I is true but Statement II is false **[Correct]**

Correct Answer: D

Solution Analysis: Both statements need to be evaluated to determine which (if any) are true: Statement (I) : SiO_2 and GeO_2 are acidic while SnO and PbO are amphoteric in nature. This statement is generally true. The oxides of silicon (SiO_2) and germanium (GeO_2) are indeed acidic. These elements belong to group 14 of the periodic table and as you move down the group, the acidic character of the oxides decreases while the basic character increases. Tin oxide (SnO) and lead oxide (PbO) are lower in the group and they are amphoteric, which means they show both acidic and basic properties.